LASER Marking Training Material - Ather

Prepared By
Technical Training Team

What is LASER?

- ✓ Light Amplification by Stimulated Emission of Radiation

Main Process

- ✓ To print **barcode** at both side of PCBA

Main Board

- ❖ Top Side – 1 barcode called SN(Serial Number)
- ❖ Bottom Side – 2/4 barcode called(Part/Product serial Number)

Models :

Dashboard	- 2 & 3
Medulla	- 2 & 4
NIPDU	- 4
BMS	- 1 &2
Drive Power	- 1
Control board	- 2

Stages:

PCB Loading

↓

SN Printing

↓

Flipper

↓

PSN printing

↓

PCB Unloading

Equipment Used:

I Share

Sonic

HEAD PICTURES

Sonic

I Share

PCB Loader

SN printing

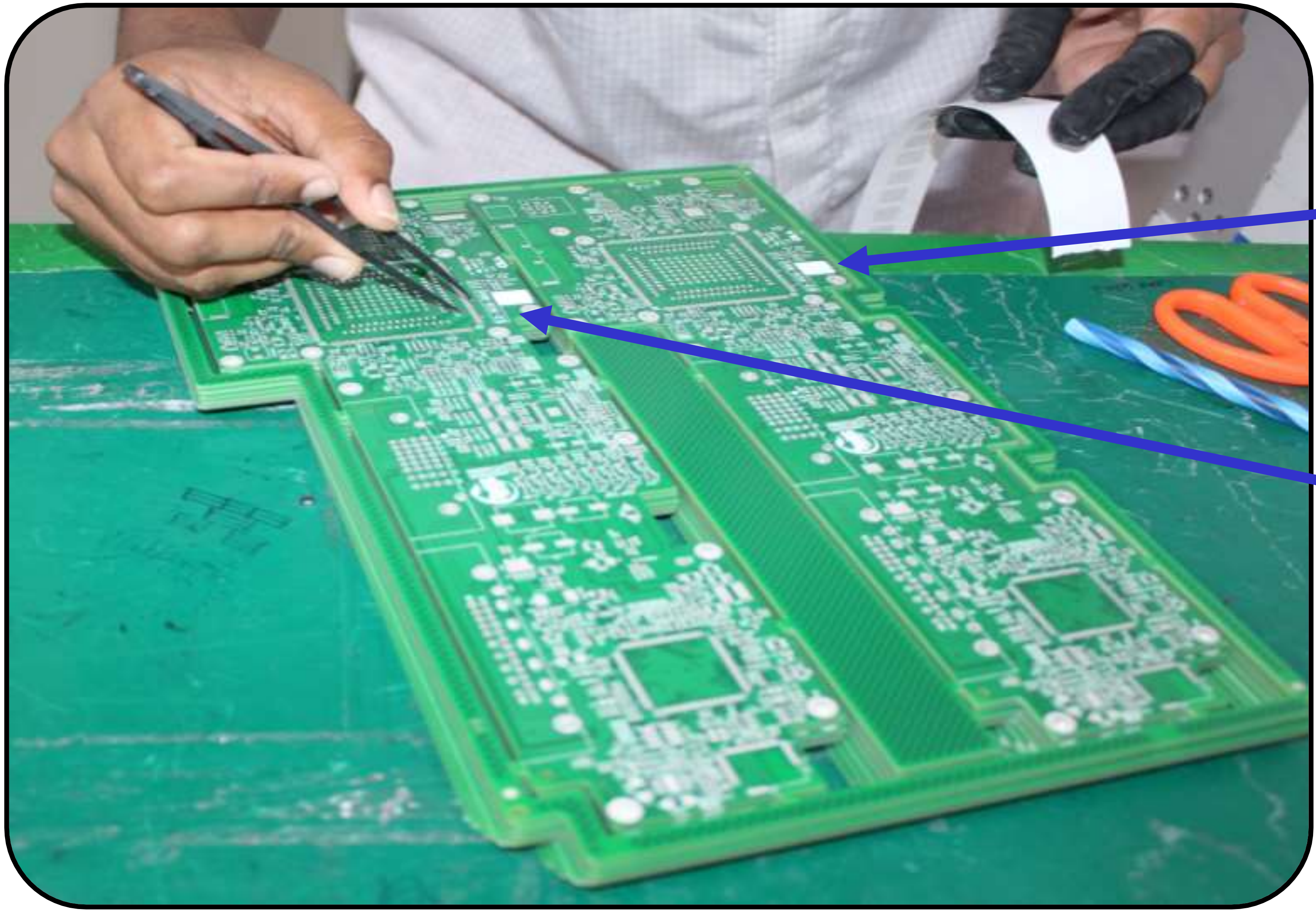
Flipper

PSN printing

Unloading

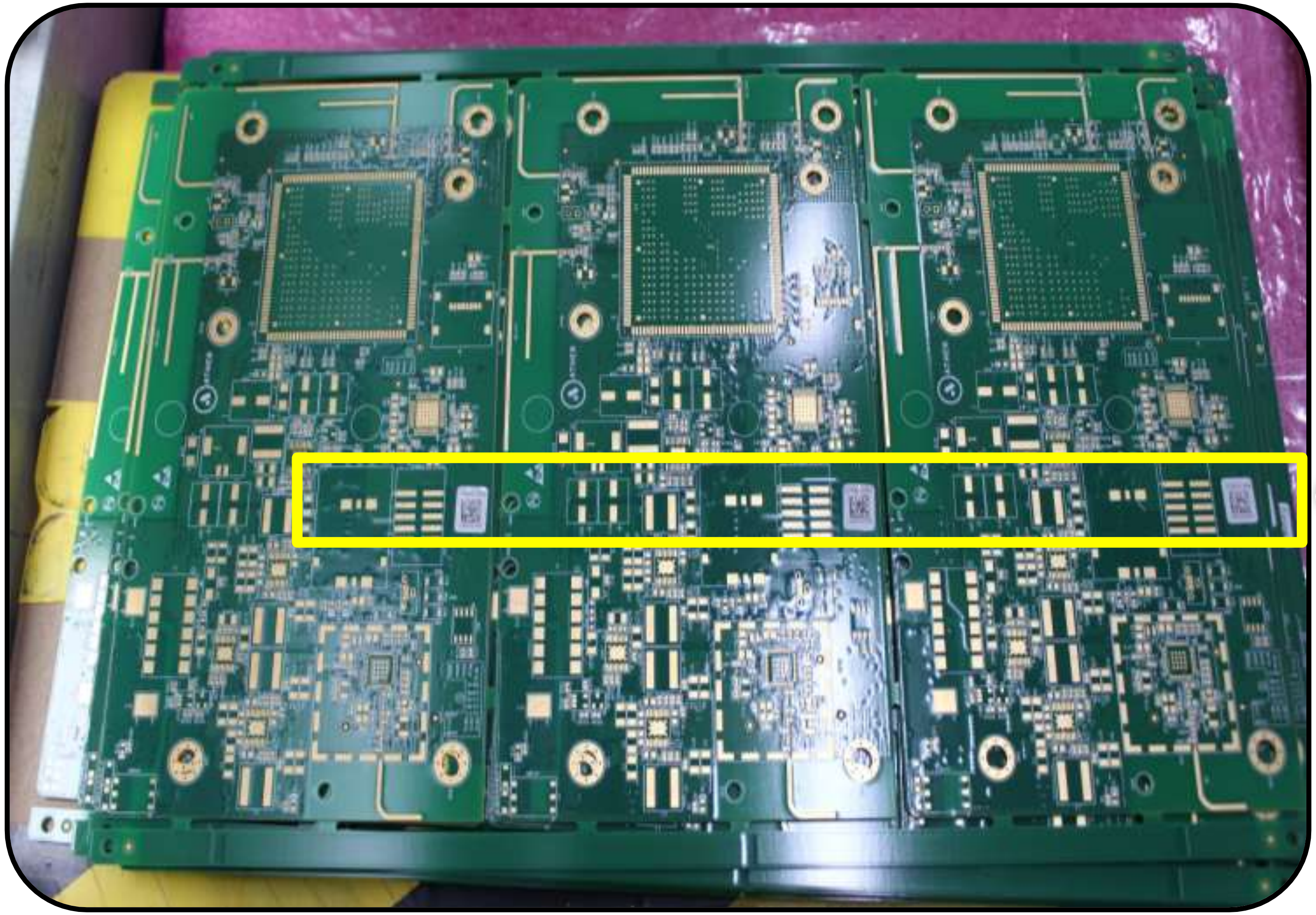
Storage condition :

Label pasting process on PCB

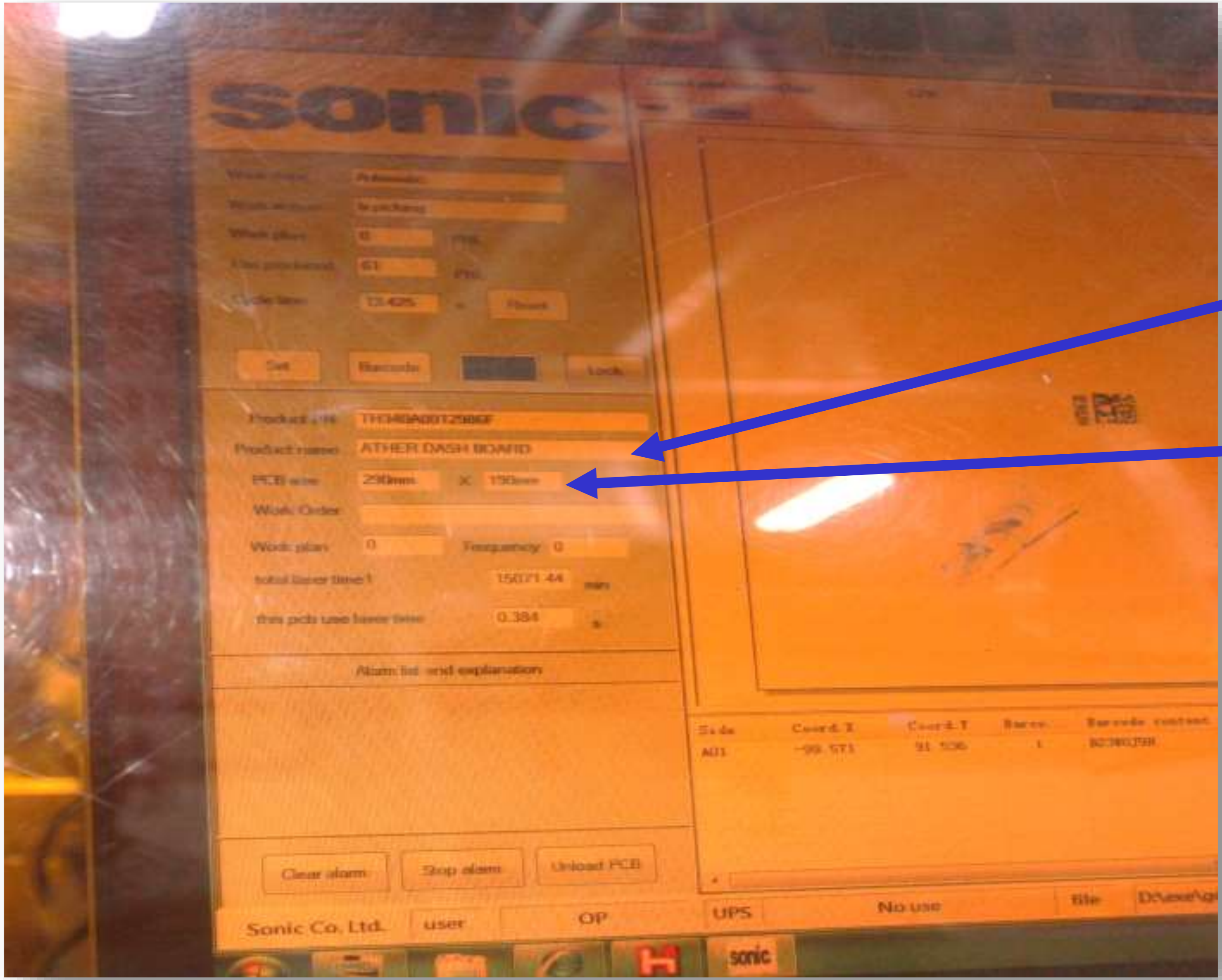


Label pasting location 1

Label pasting location 2

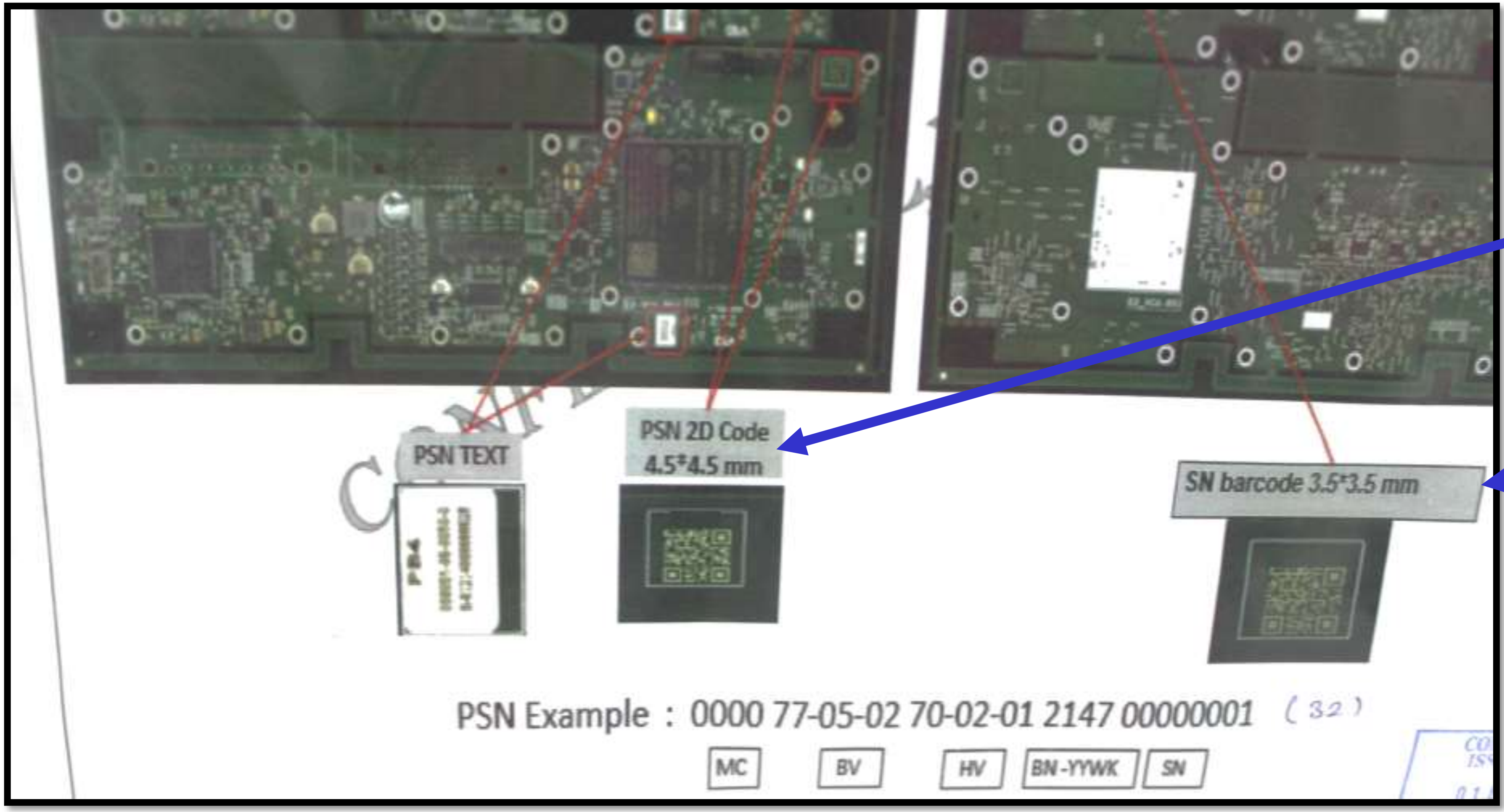


Barcode location & size



Model name

PCB size



PSN Size

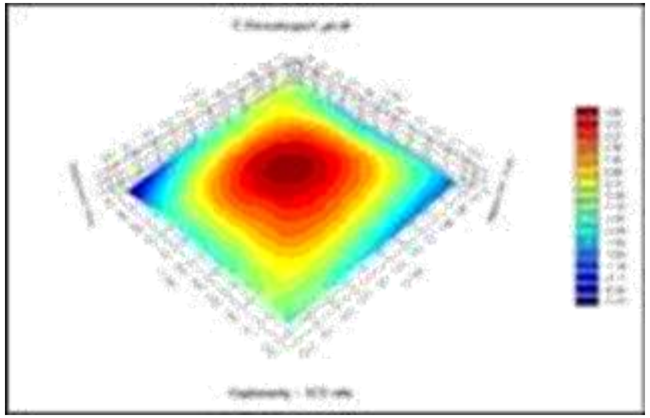
SN Size

SN & PSN Size mentioned in SOP

MSD – Introduction

- In a standard SMT assembly process there exist two major types of workmanship-related component defects that can escape all existing inspection and test processes, and result in early life failure of systems in the field. The first type is related to the Electro Static Discharge (ESD) sensitivity of the electronic circuit on the semiconductor. In general this phenomenon is well understood in the manufacturing industry and the proper systems and procedures are in place and well observed.

- The second type of damage is related to moisture-sensitive devices, which included most ICs made with plastic or organic materials. This problem has been observed and documented since the early days of SMT technology and there are industrial standards that dictate the proper procedures. However, this failure often goes undetected.



What is MSD

- **MSD – Moisture Sensitive Devices are components that absorb moisture and have a high potential for internal cracking.**
- **All the integrated circuit components are classified as a moisture sensitive device. Electronic devices encapsulated with plastic compounds and other organic materials. Moisture form atmospheric humidity enters permeable packaging materials through diffusion. The moisture preferentially collects at dissimilar material interfaces.**
- **In traditional assembly process, plated through holes assembly. It is not required the surface mount reflow.**
  - **Component is not required to solder by the high reflow temperature.**
  - **Component was soldered by wave soldering process. The component temperature is below the evaporate point.**
  - **No concern for the moisture damage.**
- **Higher density, small product size to drive the SMT. During solder reflow, the combination of rapid moisture expansion and materials mismatch can result in package cracking and/or de-lamination of critical interfaces within the package. These internal defects are nearly impossible to detect during the PCB assembly and test process.**

Moisture Sensitive Level

➢ Definition:

The MSL is an electronic standard for the time period in which a MSD can be exposed to ambient room conditions

- MSL : Moisture Sensitive Level
- MSD : Moisture Sensitive Device
- MBB : Moisture Barrier Bag
- HIC : Humidity Indicator Card
- SMD : Surface Mount Device
- IC : Integrated circuit
- MSID : Moisture Sensitive Identification

➢ Desiccant:

An absorbent material used to maintain a low relative humidity.

➢ MBB:

A bag designed to restrict the transmission of water vapor and used to pack moisture sensitive devices.

➢ MSID:

A symbol indicating that the contents are moisture sensitive.

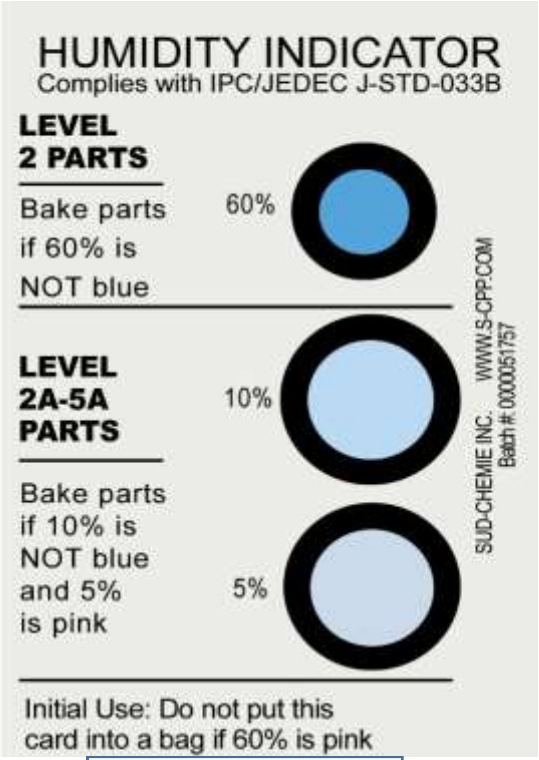
Moisture Sensitive Level

➢ <u>HIC:</u>

A card on which a moisture-sensitive chemical is applied such that it will make a significant, perceptible change in color (hue), typically from blue (dry) to pink (wet) when the indicated relative humidity is exceeded.

HIC Spot Compliance						
Spot	**Indication at 2% RH Environment**	**Indication at 5% RH Environment**	**Indication at 10% RH Environment**	**Indication at 55% RH Environment**	**Indication at 60% RH Environment**	**Indication at 65% RH Environment**
5%	Blue(dry)	Lavender (spot value) **change ≥7% hue**	Pink(Wet)	Pink(Wet)	Pink(Wet)	Pink(Wet)
10%	Blue(dry)	Blue(dry)	Lavender (spot value) **change ≥10% hue**	Pink(Wet)	Pink(Wet)	Pink(Wet)
60%	Blue(dry)	Blue(dry)	Blue(dry)	Blue(dry)	Lavender (spot value) **change ≥10% hue**	Pink(Wet)

Note: Other color schemes may be used.



HIC

Note: The HIC is packed inside the moisture-barrier bag, along with a desiccant, to aid in determining the level of moisture to which the moisture-sensitive devices have been subjected.

Moisture Sensitive Level

Moisture Sensitive level & Floor life	
Moisture Sensitivity Level	**Floor Life (out of bag) at factory ambient ≤30 °C/60% RH or as stated**
1	Unlimited at ≤30 °C/85% RH
2	1 year
2a	4 weeks
3	168 hours
4	72 hours
5	48 hours
5a	24 hours
6	Mandatory bake before use. After bake, must be reflowed within the time limit specified on the label.

HUMIDITY INDICATOR CARD

- Humidity indicator card is used to check the humidity level of the PCB…
- If the color of the card changes from blue to pink or level exit 40% then the particular PCB package can’t be used for the production line

Process Followed :

Step 1

Step 2

Step 3

Step 4

Step 5

Step 6

lASER LINES

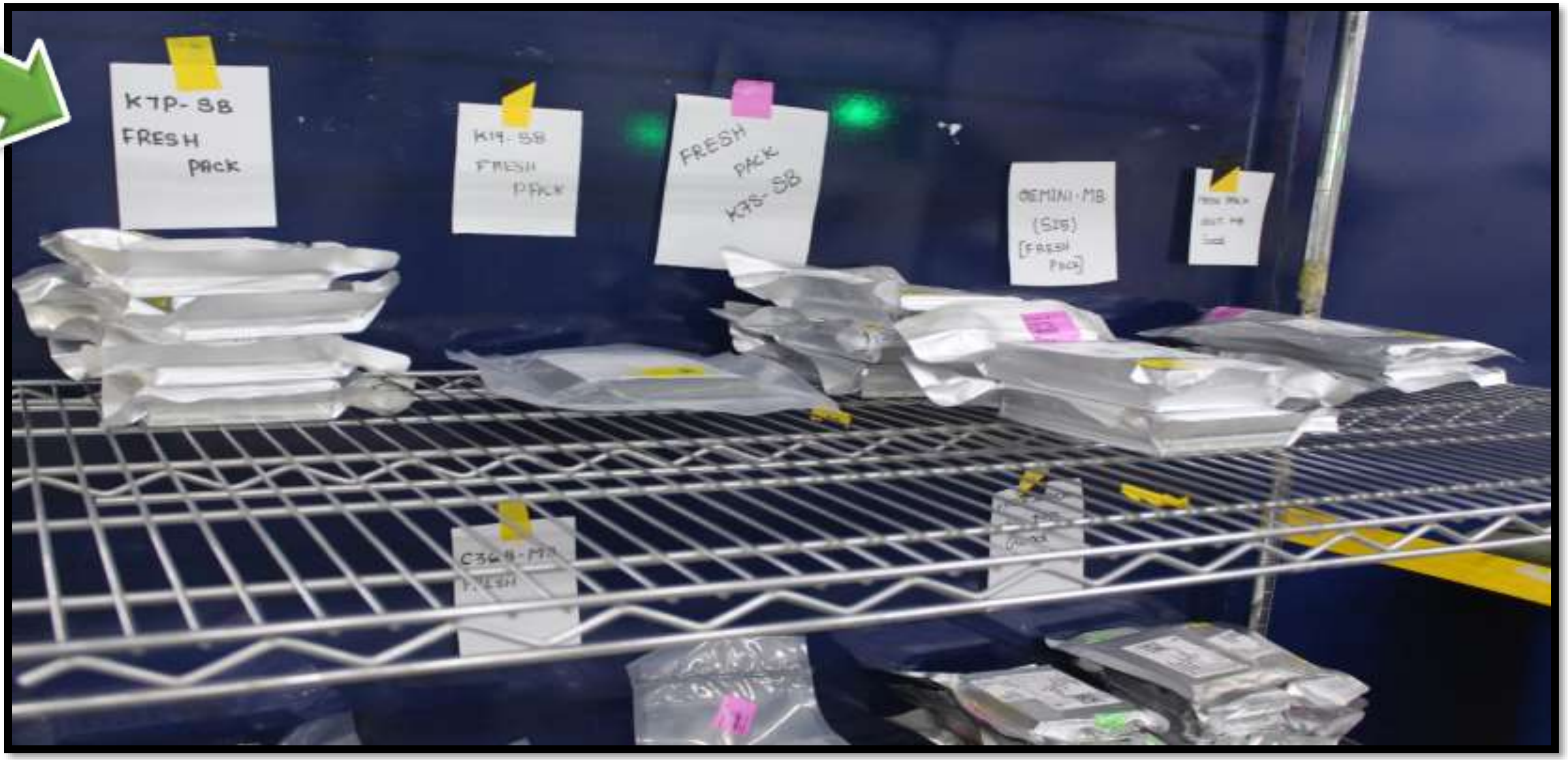
Line number mentioned in every line

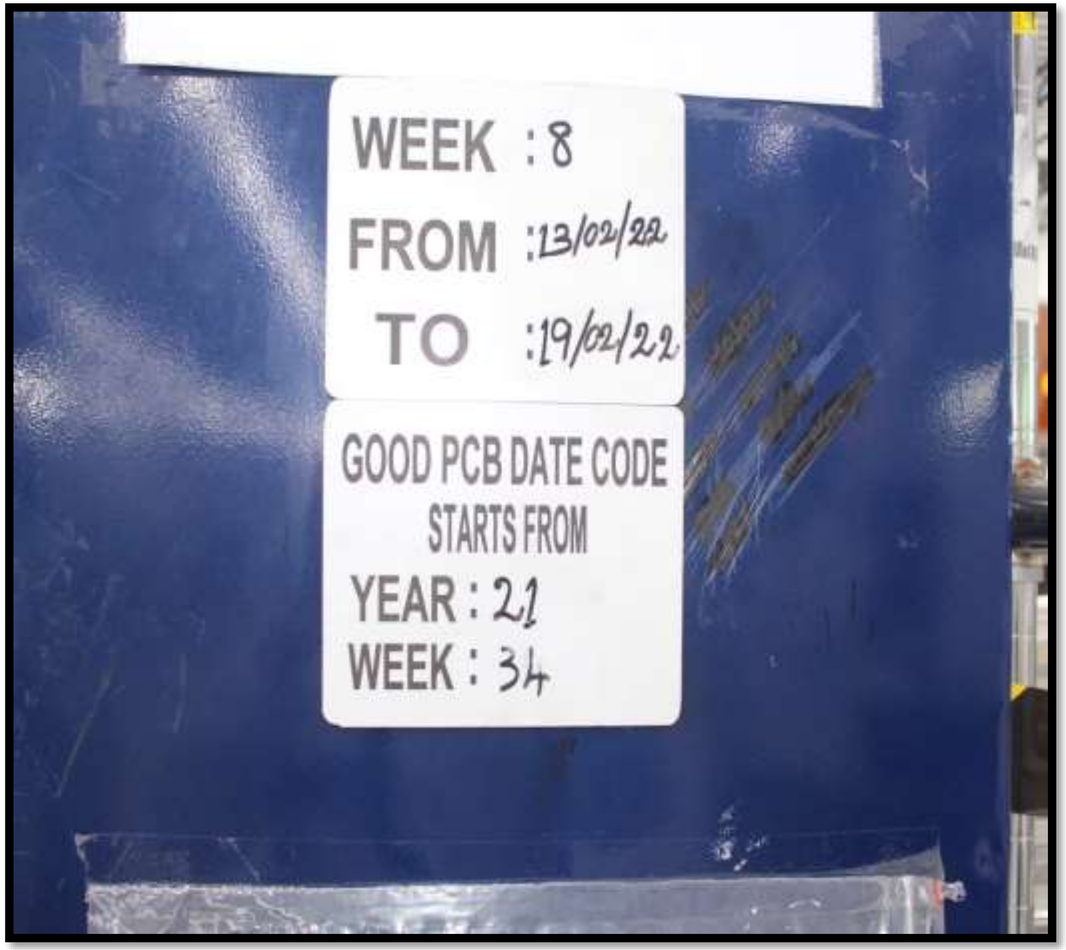
- Laser lines are sequencially located one by one
- 1st always check line and then program for the particular model

PCB received from kitting stored in trolley

- The laser operator need to collect the PCB package from the kitting and move it in a trolley with top and bottom empty .
- Ensure all the check point while receiving ex.. Part number , date code ,model , etc..

PCB received from kitting stored in trolley & model name also mentioned





Date code details: Valid up to 6 months from packing

Line number and model of the fresh PCB is mentioned

Part Number

Date code

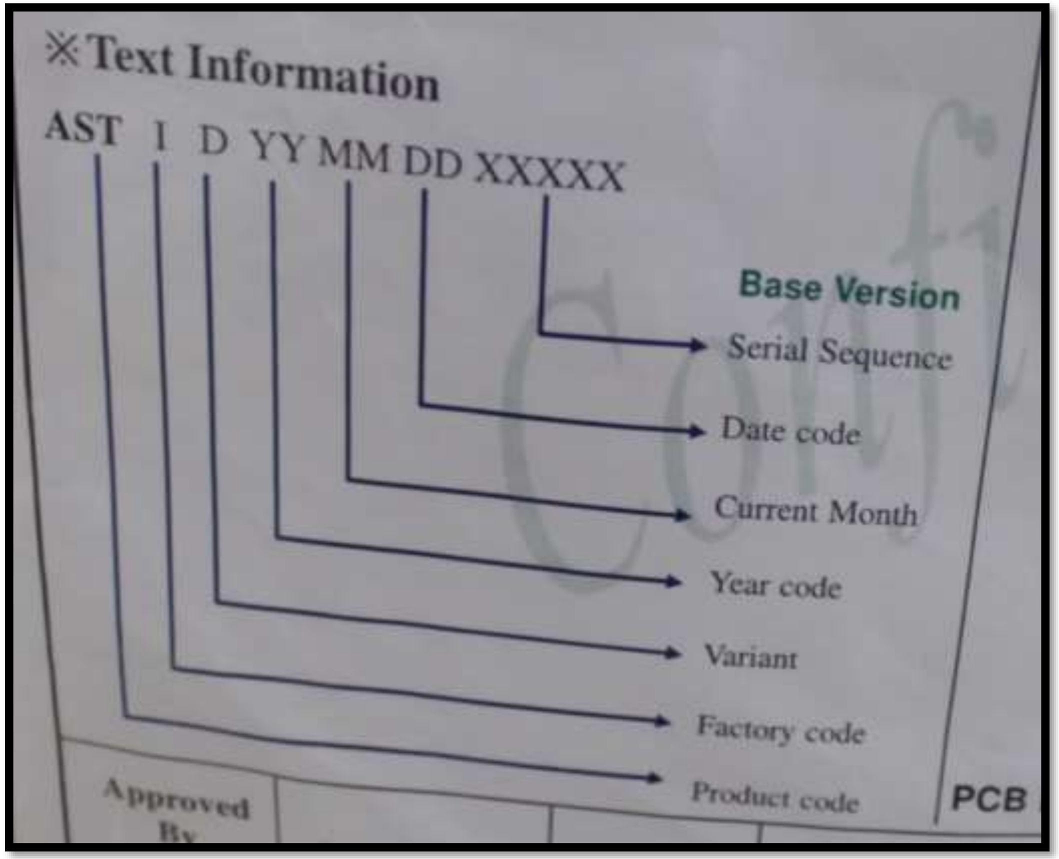
Moisture Sensitive Level

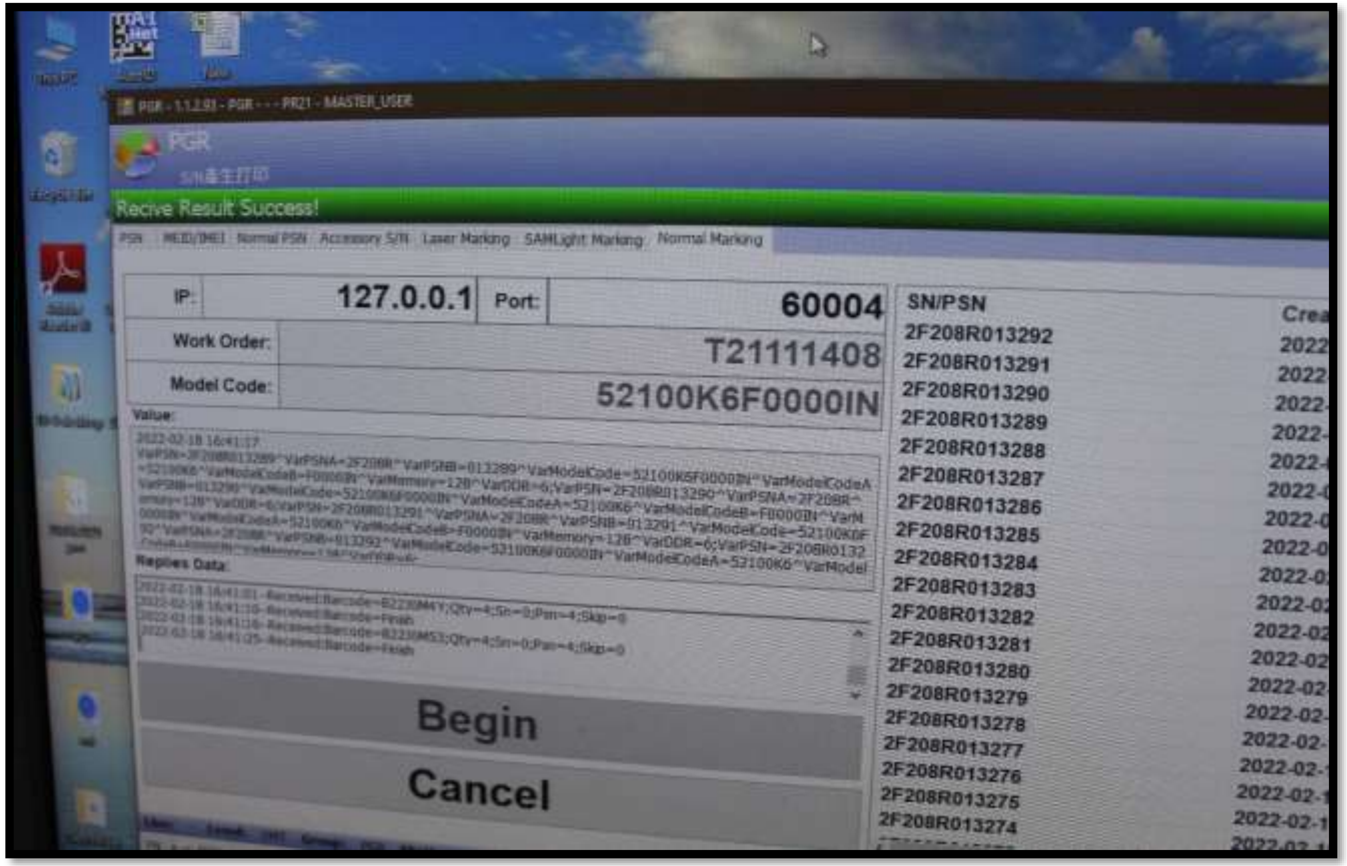
Quantity(Panel)

Restriction Of Hazardous Substance

Blue color Bin

Qubo – For Dust cleaning





SN and PSN – Every digit meaning defined in SOP

PGR software used in LASER stage

ESD Material in LASER Stage:

- Apron
- ESD Cap
- ESD slipper
- Finger Cots

Apron

Head Cap

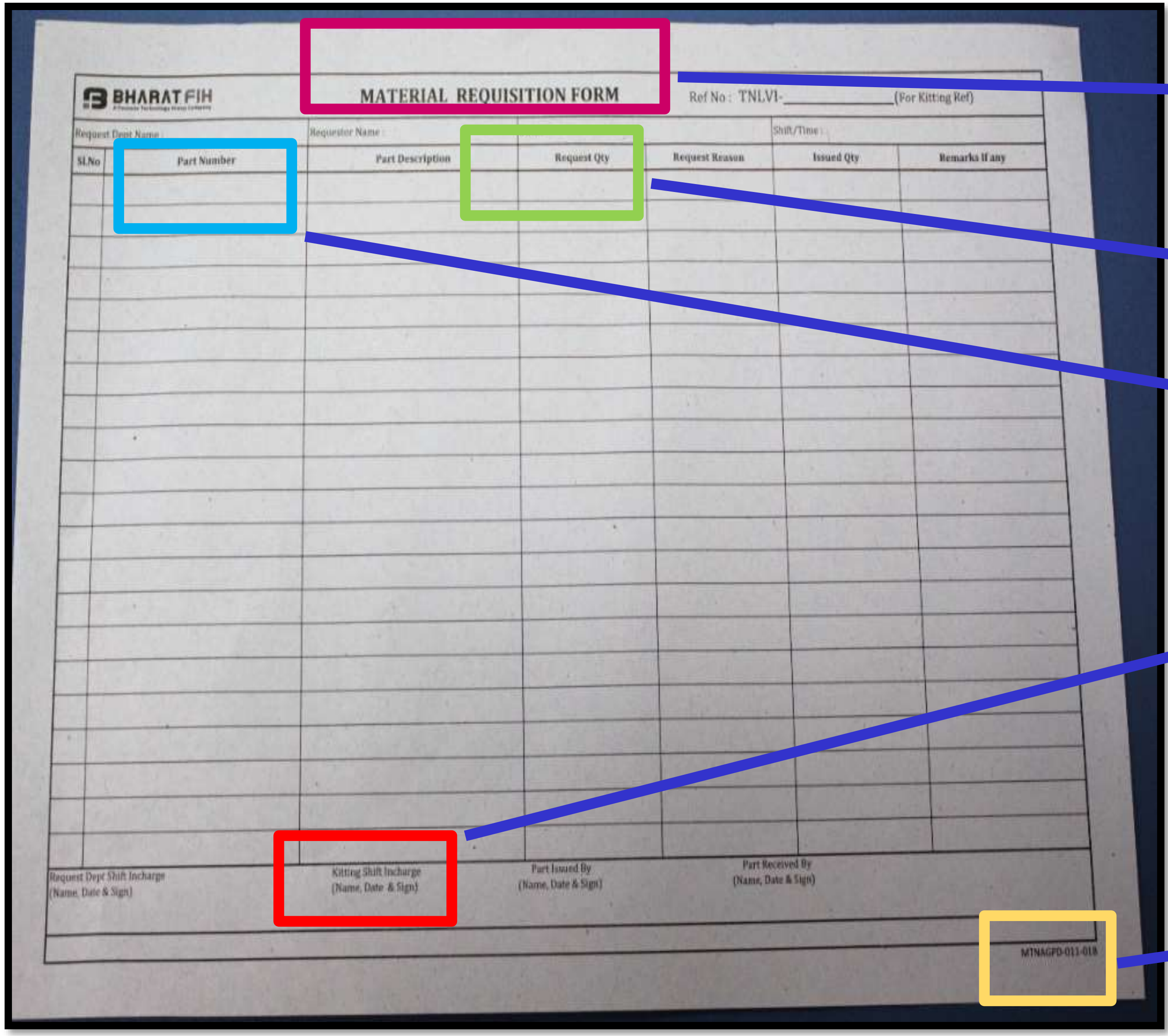
Slipper

Wrist Band

Antistatic Tools

Air ionizer

Foot strap



BHARAT FIH

MATERIAL REQUISITION FORM

Ref No : TNLVI-__________(For Kitting Ref)

Request Dept Name :

Requestor Name :

Shift/Time :

Sl.No	Part Number	Part Description	Request Qty	Request Reason	Issued Qty	Remarks If any

Request Dept Shift Incharge (Name, Date & Sign)

Kitting Shift Incharge (Name, Date & Sign)

Part Issued By (Name, Date & Sign)

Part Received By (Name, Date & Sign)

MTNAGPD-011-018

Used for material requesting to kitting

Requested Quantity

Requested part number

Kitting In charge sign

Document Number

Thank you